MCQs: Computer Networks (Class XII)

Introduction to Computer Networks

1. A group of two or more similar things or people interconnected with each other is called a
   (A) System
   (B) Server
   (C) Network
   (D) Protocol
2. Which of the following best describes today's world according to the text?
   (A) Isolated
   (B) Partially connected
   (C) Fully independent
   (D) Connected
3. Information in the modern world is produced, exchanged, and traced
   (A) Periodically
   (B) Manually
   (C) In real time
   (D) Offline
4. Which of the following is NOT mentioned as an example of a network in daily life?
   (A) Social network
   (B) Mobile network
   (C) Airline network
   (D) Weather network
5. A computer network is defined as an interconnection among
   (A) One computer only
   (B) Two or more computers or computing devices
   (C) Only servers
   (D) Only mobile phones
6. Which of the following is a benefit of a computer network?
   (A) Data isolation
   (B) Resource sharing
   (C) Reduced storage
   (D) Manual communication
7. The size of a computer network depends on
   (A) Type of cable
   (B) Number of computers connected
   (C) Operating system
   (D) Software used
8. Which of the following can act as a host or node in a network?
   (A) Desktop
   (B) Server

(C) Laptop
(D) All of the above

9. Devices like switch, router, and modem are collectively called
(A) Storage devices
(B) Input devices
(C) Networking devices
(D) Processing devices

10. Data in a network is divided into smaller chunks called
(A) Frames
(B) Blocks
(C) Packets
(D) Signals

Nodes and Communication

11. Devices in a network can be connected through
(A) Only wired media
(B) Only wireless media
(C) Wired or wireless media
(D) Optical media only

12. In a communication network, a device that can receive, create, store, or send data is called a
(A) Server
(B) Node
(C) Gateway
(D) Medium

13. Which of the following can be considered a node?
(A) Router
(B) Printer
(C) Digital telephone handset
(D) All of the above

14. Interconnectivity of devices allows information exchange through
(A) Email
(B) Websites
(C) Audio/video calls
(D) All of the above

15. Sharing a printer among multiple computers is an example of
(A) Data isolation
(B) Resource sharing
(C) Signal regeneration
(D) Encryption

16. A hotspot usually forms which type of network?
(A) WAN

(B) MAN
(C) LAN
(D) Personal network

Evolution of Networking

17. ARPANET was commissioned in the
    (A) 1950s
    (B) 1960s
    (C) 1970s
    (D) 1980s

18. ARPANET was commissioned by
    (A) NASA
    (B) CERN
    (C) U.S. Department of Defence
    (D) NSF

19. The first ARPANET message was sent between
    (A) MIT and Harvard
    (B) UCLA and SRI
    (C) CERN and NSF
    (D) IBM and Intel

20. The term "Internet" was coined in
    (A) 1969
    (B) 1971
    (C) 1974
    (D) 1983

21. TCP/IP was introduced as a standard protocol on ARPANET in
    (A) 1961
    (B) 1971
    (C) 1974
    (D) 1983

22. Who developed network messaging or E-mail?
    (A) Tim Berners-Lee
    (B) Roy Tomlinson
    (C) Vint Cerf
    (D) Stewart Kirkpatrick

23. The symbol "@" in email means
    (A) Hash
    (B) Dot
    (C) At
    (D) Mail

24. NSFNET program was initiated by
    (A) CERN

(B) ARPANET
(C) National Science Foundation
(D) IANA

25. The first version of Wi-Fi (802.11) was introduced in
(A) 1986
(B) 1990
(C) 1997
(D) 2000

26. HTML and URL were developed at
(A) MIT
(B) CERN
(C) UCLA
(D) NSF

Types of Networks

27. Computer networks are categorised based on
(A) Cost
(B) Operating system
(C) Geographical area and data transfer rate
(D) Device brand

28. PAN stands for
(A) Public Area Network
(B) Personal Area Network
(C) Private Access Network
(D) Peripheral Area Network

29. LAN stands for
(A) Local Access Network
(B) Logical Area Network
(C) Local Area Network
(D) Long Area Network

30. MAN stands for
(A) Major Area Network
(B) Metropolitan Area Network
(C) Medium Area Network
(D) Managed Area Network

31. WAN stands for
(A) Wireless Area Network
(B) Web Area Network
(C) Wide Area Network
(D) Wired Access Network

Personal Area Network (PAN)

32. PAN typically covers a range of about
    (A) 1 metre
    (B) 5 metres
    (C) 10 metres
    (D) 100 metres

33. Which of the following can form a wired PAN?
    (A) Bluetooth
    (B) USB connection
    (C) Wi-Fi
    (D) Ethernet backbone

34. Bluetooth communication between smartphones forms
    (A) LAN
    (B) WAN
    (C) Wired PAN
    (D) Wireless PAN

Local Area Network (LAN)

35. LAN covers
    (A) A city
    (B) A country
    (C) A limited geographical area
    (D) The entire world

36. LAN connectivity can be achieved using
    (A) Ethernet cables
    (B) Fibre optics
    (C) Wi-Fi
    (D) All of the above

37. LAN is comparatively secure because
    (A) It uses encryption
    (B) Only authentic users can access resources
    (C) It is wireless
    (D) It is slower

38. Maximum extension of LAN is about
    (A) 100 m
    (B) 500 m
    (C) 1 km
    (D) 10 km

39. Data transfer rate in LAN ranges from
    (A) 1–5 Mbps
    (B) 5–10 Mbps
    (C) 10–1000 Mbps
    (D) 1–10 Gbps

40. Ethernet refers to
    (A) A cable
    (B) A device
    (C) A set of rules
    (D) A protocol for WAN

Metropolitan Area Network (MAN)

41. MAN covers a
    (A) Room
    (B) Building
    (C) City or town
    (D) Country

42. Data transfer rate in MAN is
    (A) Higher than LAN
    (B) Equal to LAN
    (C) Lower than LAN
    (D) Zero

43. Cable TV network is an example of
    (A) PAN
    (B) LAN
    (C) MAN
    (D) WAN

44. MAN can extend up to
    (A) 1 km
    (B) 10 km
    (C) 30–40 km
    (D) 100 km

Wide Area Network (WAN)

45. WAN connects
    (A) Only computers
    (B) Only servers
    (C) LANs and MANs across large distances
    (D) Only mobile phones

46. The Internet is an example of
    (A) PAN
    (B) LAN
    (C) MAN
    (D) WAN

47. WANs are commonly used by
    (A) Home users only
    (B) Large organisations

(C) Single room networks
(D) Bluetooth devices

Network Devices

48. Which of the following is NOT a networking device?
    (A) Router
    (B) Modem
    (C) Repeater
    (D) Monitor

49. Modem stands for
    (A) Mode Device
    (B) Modulator Demodulator
    (C) Modern Device
    (D) Module Decoder

50. The modem at the sender's end performs
    (A) Demodulation
    (B) Amplification
    (C) Modulation
    (D) Filtering

51. The modem at the receiver's end performs
    (A) Modulation
    (B) Demodulation
    (C) Routing
    (D) Switching

Ethernet Card / NIC

52. NIC stands for
    (A) Network Internet Card
    (B) Network Interface Card
    (C) Node Interface Card
    (D) Network Input Card

53. Ethernet card is mounted on the
    (A) RAM
    (B) Hard disk
    (C) Motherboard
    (D) Power supply

54. Each NIC has a unique
    (A) IP address
    (B) Port number
    (C) MAC address
    (D) URL

RJ45

55. RJ45 is an
    (A) Optical connector
    (B) Eight-pin connector
    (C) Wireless adapter
    (D) Software interface

56. RJ45 is used with
    (A) Fibre optic cables
    (B) Coaxial cables
    (C) Ethernet cables
    (D) USB cables

Repeater

57. Signals usually weaken after travelling about
    (A) 10 m
    (B) 50 m
    (C) 100 m
    (D) 1 km

58. A repeater is an
    (A) Digital device
    (B) Analog device
    (C) Storage device
    (D) Input device

59. The main function of a repeater is to
    (A) Route data
    (B) Regenerate signals
    (C) Encrypt data
    (D) Filter packets

Hub and Switch

60. A hub sends incoming data
    (A) To selected devices only
    (B) To the router
    (C) To all connected devices
    (D) To the gateway

61. Data collision occurs in
    (A) Switch
    (B) Router
    (C) Hub
    (D) Gateway

62. A switch forwards data based on
    (A) IP address only
    (B) Destination address in packet
    (C) Signal strength
    (D) Cable length

63. A switch drops packets that are
    (A) Small
    (B) Large
    (C) Noisy or corrupted
    (D) Encrypted

Router

64. A router connects
    (A) Two computers
    (B) LAN to Internet
    (C) Printer to PC
    (D) Keyboard to CPU

65. Routers can
    (A) Analyse data
    (B) Repackage packets
    (C) Transmit to other networks
    (D) All of the above

66. Home Wi-Fi routers often act as
    (A) Only switches
    (B) Only modems
    (C) Router + modem/switch
    (D) Only repeaters

Gateway

67. Gateway acts as
    (A) Storage point
    (B) Entry and exit point of a network
    (C) Signal booster
    (D) Encryption device

68. All incoming and outgoing data must pass through
    (A) Switch
    (B) Hub
    (C) Gateway
    (D) NIC

69. A gateway can be implemented using
    (A) Software only
    (B) Hardware only

(C) Both software and hardware
(D) None of these

70. Firewalls are usually integrated with
    (A) Switch
    (B) Hub
    (C) Gateway
    (D) Repeater

Networking Topologies

71. Network topology refers to
    (A) Speed of network
    (B) Cost of network
    (C) Arrangement of devices
    (D) Size of network

72. Which is NOT a common topology?
    (A) Mesh
    (B) Ring
    (C) Star
    (D) Linear

Mesh Topology

73. In mesh topology, each device is connected to
    (A) Central device
    (B) Two devices
    (C) Every other device
    (D) Backbone cable

74. Mesh topology is more secure because
    (A) Data is encrypted
    (B) Each cable carries different data
    (C) It is wireless
    (D) It is slow

75. The major disadvantage of mesh topology is
    (A) Low speed
    (B) High cabling cost
    (C) Poor reliability
    (D) Central failure

Ring Topology

76. In ring topology, each node is connected to
    (A) One node
    (B) Two nodes

(C) Three nodes
(D) All nodes

77. Data transmission in ring topology is
(A) Bidirectional
(B) Multidirectional
(C) Unidirectional
(D) Random

Bus Topology

78. Bus topology uses a
(A) Central hub
(B) Central switch
(C) Single backbone cable
(D) Router

79. Bus topology is cheaper because
(A) Uses fewer cables
(B) Uses fibre optics
(C) Uses wireless links
(D) Has no nodes

Star Topology

80. In star topology, devices are connected to
(A) Bus
(B) Ring
(C) Central node
(D) Gateway

81. Failure of central device in star topology results in
(A) Partial failure
(B) No effect
(C) Complete network failure
(D) Improved speed

Tree / Hybrid Topology

82. Tree topology is also known as
(A) Linear topology
(B) Hybrid topology
(C) Circular topology
(D) Random topology

83. Tree topology is commonly used in
(A) PAN
(B) LAN

(C) MAN
(D) WAN

MAC Address

84. MAC stands for
    (A) Machine Access Code
    (B) Media Access Control
    (C) Multiple Access Channel
    (D) Memory Access Control

85. MAC address is
    (A) Temporary
    (B) Logical
    (C) Permanent
    (D) Changeable

86. MAC address length is
    (A) 32 bits
    (B) 48 bits
    (C) 64 bits
    (D) 128 bits

87. The first 24 bits of MAC address represent
    (A) Serial number
    (B) Device ID
    (C) OUI
    (D) IP prefix

IP Address

88. IP stands for
    (A) Internet Program
    (B) Internal Path
    (C) Internet Protocol
    (D) Interface Protocol

89. IPv4 address length is
    (A) 16 bits
    (B) 32 bits
    (C) 48 bits
    (D) 128 bits

90. IPv4 is written as
    (A) Hexadecimal groups
    (B) Binary strings
    (C) Four decimal numbers separated by dots
    (D) Eight decimal numbers

91. IPv6 address length is
    (A) 32 bits
    (B) 64 bits
    (C) 96 bits
    (D) 128 bits
92. IPv6 is written in
    (A) Decimal
    (B) Binary
    (C) Hexadecimal
    (D) Octal

Internet and WWW

93. The Internet is a
    (A) Software
    (B) Protocol
    (C) Global network
    (D) Browser
94. Smart devices like ACs and refrigerators are part of
    (A) LAN
    (B) IoT
    (C) MAN
    (D) PAN
95. WWW was invented in
    (A) 1986
    (B) 1990
    (C) 1997
    (D) 2000
96. HTML is used to
    (A) Route data
    (B) Design web pages
    (C) Encrypt content
    (D) Resolve domain names
97. HTTP is used to
    (A) Store data
    (B) Retrieve web pages
    (C) Assign IPs
    (D) Connect cables
98. The secure version of HTTP is
    (A) FTP
    (B) TCP
    (C) HTTPS
    (D) SMTP

Domain Name System (DNS)

99. Domain names are assigned because
    (A) IPs are slow
    (B) IPs are difficult to remember
    (C) IPs are temporary
    (D) IPs are encrypted

100. Domain name resolution means
    (A) Assigning MAC address
    (B) Encrypting data
    (C) Converting domain name to IP address
    (D) Compressing packets

101. DNS server is used to
    (A) Store web pages
    (B) Resolve domain names
    (C) Encrypt data
    (D) Route packets

102. Root DNS servers are
    (A) 10
    (B) 11
    (C) 12
    (D) 13

103. DNS root servers are named from
    (A) A–Z
    (B) A–M
    (C) A–Z excluding vowels
    (D) A–L

104. DNS root server list is maintained by
    (A) CERN
    (B) ARPANET
    (C) IANA
    (D) ISP